

URITECT

Renal Follow-up

Implementation Comparison and Sign-off

Version 1.1

Rule set	renal_followup_rules_v1.1_20260919
Android app	Uritect 1.3.0 (versionCode 4)
Application ID	ph.edu.wvsu.uritect
Strip profile	URS-10T protein/blood at 60 seconds
Purpose	Final physician comparison of the specification and implemented behavior
APK SHA-256	F68A88D576BCF34FBB83502564512E86F4E28C0CEA71C4FAF752427EBF4F975E

This packet identifies the exact tested release and exposes all implemented renal follow-up rules for final clinical comparison. It does not claim diagnostic validation.

Prepared for final physician review and documented sign-off

Clinical specification: URIRECT Renal-Related Follow-up Rule Specification v1.1 Implemented rule set: renal_followup_rules_v1.1_20260919 Message version: renal_messages_v1.1_20260919 Android app version: 1.3.0 (versionCode 4) Android application ID: ph.edu.wvsu.uritect Production visual model: production_semiquant_knn_markerless_roi_topfix_v3_20260908 Supported software strip profile: urs10t_protein_blood_60s_v1 Date prepared: 20 September 2026

1. Purpose of This Review

This packet allows the reviewing physician to compare the physician-reviewed renal follow-up specification with the exact software implementation included in the release APK.

The module recommends follow-up action. It does not diagnose kidney disease, calculate renal-disease probability, stage chronic kidney disease, or confirm hematuria.

2. Frozen Artifact Identity

Artifact	SHA-256
Release APK Uritect_v1.3.0_bayesian_v1.1_renal_v1.1_release.apk	F68A88D576BCF34FBB83502564512E86F4E28C0CEA71C4FAF752427EBF4F975E
Implemented engine renal_followup.dart	11260233ABCDD9886733F86DFD03A1C0484DE8ED849F5A245846A55EEE072CE
Clinical specification v1.1	8B462633B3B47E0C3877A4133A8A7600892B18B464100CF36C9BC765809E173A

Any change to these files changes the checksum and requires a new implementation comparison.

3. Final Action Priority

PROMPT_CONSULT > RETAKE > CONSULT > REPEAT_CONFIRM > OBSERVE

Every matching rule is stored in the local audit record. The app displays the action with the highest priority.

An invalid scan does not suppress serious symptom advice. When an invalid scan and a safety rule occur together, PROMPT_CONSULT is displayed and no analyte value is interpreted.

4. URS-10T Protein and Blood Mapping

Item	Exact implementation
Protein read time	60 seconds
Blood read time	60 seconds
Protein negative	Neg, Negative
Protein trace	Trace
Protein positive	0.3, 1.0, 3.0, >=20.0 g/L
Blood negative	Neg, Negative
Blood positive	Non-hemolyzed 10, Hemolyzed 10, Small 25, Moderate 80, Large 200 cells/uL
Combined-rule protein threshold	0.3 g/L or higher; trace is excluded

Physical Product Confirmation Required

URS-10T is not a unique manufacturer identifier. Current dataset metadata contains GenericStrip, ModelX, and LOT-TBD. Before external release, complete these fields directly from the physical bottle, box, and IFU used for final testing:

- Manufacturer: _____
- Legal manufacturer address: _____

- Product/catalog number: _____
- Lot number: _____
- Expiration date: _____
- IFU document identifier and revision: _____
- Protein reaction time printed on product chart: _____
- Blood reaction time printed on product chart: _____
- Does the physical chart exactly match Section 4? [] Yes [] No

If the physical chart does not match, the current strip profile must not be approved.

5. Implemented Checklist Inputs

Urinary, Alternate-cause, and Systemic Inputs

- Dysuria or burning during urination
- Urinary frequency
- Urinary urgency
- Lower abdominal or suprapubic pain
- Visible blood in urine
- Vaginal discharge
- Vaginal irritation
- Back or flank pain
- Fever or chills
- Nausea or vomiting

Renal Follow-up and Safety Inputs

- Blood clots in urine
- Difficulty or inability to urinate
- Markedly reduced or absent urine output
- Edema or unusual swelling
- Persistent or unexplained swelling
- Shortness of breath
- Vomiting prevents fluids or oral medication
- Confusion, fainting, or severe weakness
- Severe flank or abdominal pain

Interference and Temporary-cause Inputs

- Menstruation or vaginal bleeding
- Possible specimen contamination
- Recent strenuous exercise
- Possible dehydration
- Current acute illness
- Possible or clinically managed UTI

Repeat-test and Technical Inputs

- Protein remained present on a properly collected repeat test
- Blood remained present on a properly collected repeat test
- Blood was confirmed by urine microscopy
- A previous protein finding became negative on repeat
- Strip was expired
- Strip or reagent pads were damaged
- Protein or blood was scanned outside 60 seconds
- Strip was not the supported URS-10T profile

The possible/managed UTI input is entered explicitly. It is not inferred from the Bayesian UTI percentage. Persistence is entered explicitly and is never inferred from one scan.

6. Exact Implemented Rule Comparison

Technical Rules

ID	Implemented condition	Action
TECH-01	Invalid scan, fewer than ten result rows, unknown protein category, or unknown blood category	RETAKE
TECH-02	Expired strip, damaged strip, outside 60-second timing, or unsupported strip profile	RETAKE

Safety Rules

ID	Implemented condition	Action
SAFE-01	Visible blood with clots or inability to urinate	PROMPT_CONSULT
SAFE-02	Markedly reduced or absent urine output	PROMPT_CONSULT
SAFE-03	Edema with shortness of breath	PROMPT_CONSULT
SAFE-04	Fever/chills with flank or back pain	PROMPT_CONSULT
SAFE-05	Fever or flank/back pain with nausea/vomiting	PROMPT_CONSULT
SAFE-06	Nausea/vomiting plus inability to maintain fluids or oral medication	PROMPT_CONSULT
SAFE-07	Severe flank/abdominal pain with visible or reliable dipstick blood	PROMPT_CONSULT
SAFE-08	Confusion, fainting, or severe weakness with a urinary finding	PROMPT_CONSULT

Interference Rules

ID	Implemented condition	Action
INT-01	Menstruation/vaginal bleeding or possible contamination with reliable positive dipstick blood	REPEAT_CONFIRM
INT-02	Exercise, fever, dehydration, or acute illness with trace-or-higher protein	REPEAT_CONFIRM
INT-03	Explicit possible/managed UTI with trace-or-higher protein	CONSULT

Protein Rules

ID	Implemented condition	Action
PRO-01	Reliable negative protein and blood; no edema, visible blood, or safety trigger	OBSERVE
PRO-02	Trace protein; negative blood; no edema or safety trigger	REPEAT_CONFIRM
PRO-03	Protein 0.3 g/L or higher	CONSULT
PRO-04	User explicitly confirms protein remains present on a properly collected repeat	CONSULT
PRO-05	Protein is currently negative and a previous positive became negative on repeat, without warning findings	OBSERVE

Blood Rules

ID	Implemented condition	Action
BLD-01	Reliable non-negative dipstick blood without visible blood or safety trigger	REPEAT_CONFIRM
BLD-02	Visible blood without a safety-rule combination	CONSULT
BLD-03	User confirms blood persists on repeat or was confirmed by microscopy	CONSULT

Combined and Edema Rules

ID	Implemented condition	Action
COMB-01	Protein 0.3 g/L or higher plus reliable non-negative dipstick blood	CONSULT
COMB-02	Protein 0.3 g/L or higher plus edema	CONSULT
COMB-03	Persistent or unexplained edema explicitly reported	CONSULT

7. User-facing Action Messages

OBSERVE - Observe for symptoms

No renal-related follow-up trigger was identified from the available reliable results and reported symptoms. This does not rule out a kidney or urinary tract condition. Consult a healthcare professional if symptoms appear, continue, or worsen.

REPEAT_CONFIRM - Repeat or confirm the test

A finding was detected that may be affected by sample collection or a temporary condition. A properly collected repeat urine test or professional consultation for confirmatory testing may be appropriate.

CONSULT - Consultation suggested

A urine-strip finding or related symptom was reported. These findings can have several causes and cannot diagnose kidney disease. Consultation with a healthcare professional and confirmatory laboratory testing are suggested.

PROMPT_CONSULT - Seek prompt medical consultation

A warning symptom or finding was reported. Please seek prompt medical consultation. If you feel severely unwell or symptoms rapidly worsen, seek emergency medical assistance. This result is not a diagnosis.

RETAKE - Unable to interpret; retake the scan

The relevant urine-strip result could not be interpreted reliably. Repeat the test using the supported strip, collection method, timing, and scanning procedure. Seek professional care based on symptoms even if scanning is unsuccessful.

8. Verification Evidence

Verification	Result
Specification rule IDs represented in code	24 of 24
Specification rule IDs represented in tests	24 of 24
Focused renal tests	31 passed
Complete Flutter test suite	44 passed
Invalid scan plus serious symptoms	Passed; safety action overrides retake
Static analysis	No issues found
Android release build	Successful

9. Physician Final Comparison Checklist

Rate each statement as Agree, Revise, or Not applicable.

Item	Agree	Revise	N/A	Required change or comment
Intended use and non-diagnostic limitation are accurate	☐	☐	☐	
Final action priority is clinically appropriate	☐	☐	☐	
Safety-rule combinations are implemented as approved	☐	☐	☐	
Invalid-scan safety behavior is appropriate	☐	☐	☐	
Interference and temporary-cause rules are appropriate	☐	☐	☐	
Protein trace and positive thresholds are appropriate for the confirmed strip	☐	☐	☐	
Dipstick blood is not presented as confirmed hematuria	☐	☐	☐	
Persistence requires an explicit repeat result	☐	☐	☐	
Possible/managed UTI is not inferred from Bayesian output	☐	☐	☐	
Combined protein, blood, and edema rules are appropriate	☐	☐	☐	
User-facing action messages are clear and safe	☐	☐	☐	
The separate renal results screen avoids diagnosis and risk-probability claims	☐	☐	☐	
The implemented v1.1 behavior matches the physician-reviewed specification	☐	☐	☐	

10. Final Decision and Sign-off

- ☐ Approved as implemented for the stated thesis research purpose.
- ☐ Approved after the documented minor revisions are completed.
- ☐ Major revision and repeat comparison required.
- ☐ Not approved for the stated use.

Conditions or required revisions:

Physician name: _____

Credentials and license number: _____

Institution: _____

Signature: _____ Date: _____

Exact specification reviewed: URITECT Renal-Related Follow-up Rule Specification v1.1

Exact app reviewed: Uritect 1.3.0 (versionCode 4)

APK SHA-256 verified by reviewer or project representative:

F68A88D576BCF34FBB83502564512E86F4E28C0CEA71C4FAF752427EBF4F975E